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ARTICLE

Genetic Evidence of Isolation by Distance and Impact of Impoundments on Genetic Diversity of Riverine Channel Catfish

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Abstract
This study used microsatellite loci to provide evidence of isolation by distance and an effect of artificial impoundments on the genetic diversity of Channel Catfish *Ictalurus punctatus* from the Wabash and Ohio rivers. The Wabash River is the longest free-flowing river east of the Mississippi River in the USA, and the Ohio River is highly impounded, yet both rivers support large Channel Catfish fisheries. There was a significant positive relationship between genetic differentiation and geographic distance indicating isolation by distance. Clustering with the programs, STRUCTURE and GENELAND, and principal component analysis revealed multiple genetic clusters, and several sites had results consistent with adults existing as mixtures of genetic groups. These results suggest that the rate of straying among reproductive sites or dispersal is dependent on geographic distance. Channel Catfish from the unimpounded Wabash River had higher genetic diversity (H_o and H_e) than that from the impounded Ohio River. The impoundments had discordant effects on genetic structure; two site comparisons across a dam were differentiated more strongly than would be expected by geographic distance, whereas genetic differentiation was not detected in another comparison across a dam. Thus, the effect of artificial impoundments on Channel Catfish biology may be affected by river or site-specific characteristics, and the presence of genetic differentiation may be minimized by large population sizes.

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Genetic analysis can provide valuable information about many characteristics of fish populations such as behavior, migration, and life histories (Ferguson and Danzmann 1998; Pereira et al. 2009). Information on the scale over which genetic structuring occurs can have important implications for the management of natural resources and provide important details about the populations in question (Slatkin 1993; Hellberg 1995; Pogson et al. 2001). Several different biological phenomena can generate genetic structure in fishes. Two common ones are isolation by distance and isolated reproductive grounds (Primmer et al. 2006; Griffiths et al. 2009; Dammannagoda et al. 2011). Isolation by distance occurs when gene flow decreases with distance among locations, which results in a positive linear relationship between genetic differentiation and geographic distance (Wright 1943; Kimura and Weiss 1964; Slatkin 1993). Isolated reproductive grounds cause individuals to originate from discrete spawning populations, yet adults may exist in mixtures if their assemblage consists of more than a single reproductive unit (Shaklee et al. 1999; Carlsson et al. 2006; Boustany et al. 2008).

Patterns of genetic structure in riverine freshwater fish are also affected by barriers to migration, such as artificial impoundments. Because impoundments can inhibit movement, they can alter patterns of gene flow, thereby affecting genetic structure and genetic diversity in fishes (Neraas and Spruell 2001; Yamamoto et al. 2004). Artificial impoundments not only inhibit movement, they can also change aspects of river water quality, nutrient load, and aquatic fauna assemblages (Meldgaard et al. 2003; Santucci et al. 2005; Csiki and Rhoads 2014).

Channel Catfish *Ictalurus punctatus* is one of the most important commercial and sport fishes in the Mississippi River drainage (Pitlo 1997; Hubert 1999; Michaletz and Dillard 1999). Therefore, it is important to determine their genetic structure to better understand their biology and potential management implications. Additionally, Channel Catfish are migratory and can exhibit homing tendencies; thus, catfish in impounded systems can have altered genetic diversities and structuring relative to those from free-flowing systems (Pellett et al. 1998). This is especially true because adult Channel Catfish were found to move throughout the Wisconsin River in the fall, but returned to the same spawning sites and summer home ranges as occupied the previous year (Pellett et al. 1998).

Since Channel Catfish are migratory and may return to the same spawning site, they could develop genetic structure by spawning site fidelity. Telemetry studies have demonstrated that Channel Catfish are capable of making movements well over 100 km, sometimes up to 500 km, which suggests that long-distance gene flow is possible (Lagler 1961; Dames et al. 1989; Gerhardt and Hubert 1990; Hegrenes 1992; Fago 1999). However, other studies found their movement to be highly variable, and that a majority of individuals did not move great distances; on average, they moved less than 16 km (Hegrenes 1992) and had relatively small summer home ranges (approximately 5.7 km: Wendel and Kelsch 1999). Further, tagged Channel Catfish in the Wabash River moved an average of only 8 km (Donabauer 2009). Given these variable estimates of Channel Catfish movement, genetic analysis may provide a better understanding of their long-term patterns of dispersal and migration, especially due to the high variability in their movements; these patterns may be dependent upon the system in question.

In this study, 10 microsatellite loci were screened in adult Channel Catfish to determine their genetic structure and diversity in the Wabash and Ohio rivers. Our objectives were to determine (1) whether there is genetic structure in Channel Catfish among sites and whether the structure can be predicted by geographic distance following an isolation-by-distance pattern, and (2) whether the presence of artificial impoundments acted as a barrier to Channel Catfish movement, thus serving as an impediment to gene flow further affecting their genetic structure. This study provides the first information on the genetic structure of a commercially and recreationally harvested wild population of Channel Catfish.

METHODS

Study sites.—The Wabash River flows for 810 km and has a 103,500-km^2 basin, and the Ohio River flows for 1,579 km and has a 490,601-km^2 basin. Four sites were sampled on the Wabash River and three sites on the Ohio River (Figure 1; Table 1); selected sites offered consistent river access throughout the sampling season. Boat electrofishing with DC, bank poles, hoopnets, and trot lines were used to sample adult Channel Catfish, defined as those larger than 200 mm (Vokoun and Rabeni 1999; Bodine et al. 2013). A total of 120 Channel Catfish were collected from the Wabash and Ohio rivers, and sample sizes ranged from 11 to 24 per collection site (Table 1). Fin clips were taken from each individual and stored in 95% ethanol until genetic data collection. Site location and code are given in Table 1; site codes are defined as river (i.e., OH and WA) and the nearest river kilometer (rkm) of each site. The portion of the Wabash River sampled is free of artificial impoundments (Figure 1); on the Ohio River, Smithland Lock and Dam (SM L&D) occurs between sampling sites OH_1478 and OH_1511 (site codes in Table 1), and Lock and Dam 52 (L&D 52) occurs between sampling sites OH_1511 and OH_1548. Smithland Lock and Dam (a lift-gate dam) and Lock and Dam 52 (a wicket dam) are twin systems; L&D 52 was completed in 1929 but became operational in 1928, and SM L&D began operation in 1979. On SM L&D, each lock is 30.5 m (100 ft) wide and 366 m (1,200 ft) long, whereas L&D 52 has one lock that is 33.6 m (110 ft) wide and 366 m (1,200 ft) long and the second is 183 m (600 ft) long and 33.6 m (110 ft) wide. Additionally, there are no fish ladders present on either system, so the only fish

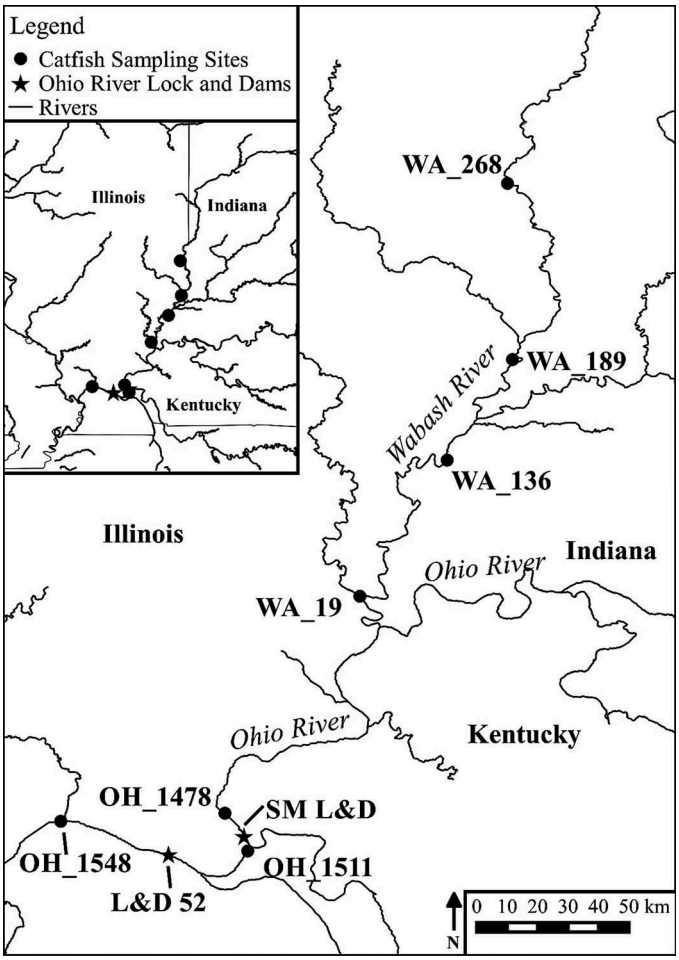

FIGURE 1. Sample sites for Channel Catfish on the Wabash and Ohio rivers, including the location of the Smithland lock and dam (SM L&D) and lock and dam 52 (L&D 52).

passage likely available is through overflow or through the locks themselves.

Genetic data collection.—Channel Catfish were screened at 10 previously developed microsatellite loci (Tan et al. 1999). The DNA was extracted with the Qiagen DNeasy Animal Tissue Kit (Qiagen, Valencia, California). Microsatellites were amplified by multiplex PCR using a common labeled M13F primer labeled with 5′ 6-FAM or 5′ HEX (following Shimizu et al. 2002). The PCR was conducted at a final volume of 10 μL, containing 1X PCR buffer (50 mM KCl, 10 mM tris HCl pH 9.0), 2 mM $MgCl_2$, 200 μM of each dNTP, 0.1 unit Taq DNA polymerase, 0.9 μM of each PCR primer, and 1–20-ng template DNA. Thermal cycles were 94°C for 2 min, then 94°C for 30 s, 50–56°C for 30 s, and 72°C for 30 s, repeated 35 times, and finally 70°C for 5 min; this was done due to the different optimization thermocycler conditions for individual microsatellite loci. Products were then sent to the Georgia Genomics Facility (University of Georgia) for electrophoresis. Lane files were analyzed and allele sizes were defined with PEAKSCANNER (Life Technologies).

Data analysis.—Each locus in each site was tested for conformation to Hardy–Weinberg equilibrium (HWE) with GENEPOP (Rousset 2008). Based on significant deviations from HWE, each locus was tested for the presence of null alleles; then the genotypes of loci were corrected for expected null alleles by using FreeNA software (Chapuis and Estoup 2007). After correcting for null alleles, allelic richness (A_R) was estimated using HP-Rare version 1.1 (Kalinowski 2005), observed (H_o) and expected (H_e) heterozygosity and the inbreeding coefficient (F_{IS}) were estimated using GENALEX-6 version 6.501 (Peakal and Smouse 2006), and linkage disequilibrium was calculated with GENEPOP (Rousset 2008).

Overall, pairwise genetic differentiation between sites and groupings based off principal components analysis (PCA) and STRUCTURE analysis was measured by calculating the Φ_{ST} estimator of F_{ST} (Peakal and Smouse 2006) using GENALEX-6 version 6.501; 9,999 permutations were used to estimate statistical significance. A Mantel's test was used to compare pairwise genetic differentiation [$\Phi_{ST}/(1 - \Phi_{ST})$] to pairwise geographic distance (measured as the difference in river kilometers between two sites) to test for isolation by distance. The computer program ML-Relate was used to estimate relatedness (r) between all individuals (Kalinowski et al. 2006). Bonferroni correction of α values were applied where appropriate.

TABLE 1. Sites sampled, site code designations, source river (WA = Wabash River, OH = Ohio River), sample size (N), and river kilometer (rkm) for adult Channel Catfish from the Wabash and Ohio rivers. Genetic diversity indices for 10 microsatellite loci include observed (H_o) and expected (H_e) heterozygosity, inbreeding coefficient (F_{IS}), allelic richness (A_R), and relatedness (r).

Site	Site code	River	N	rkm	H_o	H_e	F_{IS}	A_R	r
Hutsonville	WA_268	Wabash	23	268.80	0.699	0.691	−0.014	5.55	0.049
St. Francisville	WA_189	Wabash	17	189.90	0.712	0.741	0.044	6.02	0.056
Crawleyville	WA_136	Wabash	20	136.80	0.709	0.754	0.063	6.51	0.038
New Haven	WA_19	Wabash	24	19.30	0.700	0.721	0.008	6.45	0.044
Smithland	OH_1478	Ohio	11	1,478.20	0.650	0.658	0.009	5.59	0.039
Pool 52	OH_1511	Ohio	13	1,511.10	0.678	0.695	0.029	6.28	0.029
Pool 53	OH_1548	Ohio	12	1,548.20	0.690	0.628	−0.115	5.42	0.034

Population structuring was assessed using the Bayesian clustering program STRUCTURE version 2.3.4 (Pritchard et al. 2000; Falush et al. 2003) and GENELAND version 4.0.6 (Guillot et al. 2005). These programs were applied to determine how many genetic groups (k) were present in the data. For STRUCTURE, independent runs were conducted for k = 1–8 using 10,000 burn-in steps and 100,000 post burn-in steps. The most likely number of k was identified by the maximum estimated natural log of the probability of observing the data given the number of groups and the delta k method of Evanno et al. (2005). STRUCTURE HARVESTER was used to determine delta k for each cluster (Earl and Von Holdt 2012). Individuals were assigned to each of the k groups if the relevant q-value was greater than 0.8. For GENELAND, the spatial, uncorrelated allele frequencies model was utilized to determine the most appropriate k value. Initially, we ran 10 runs allowing k to vary from 1 to 10 with 100,000 iterations, maximum rate of Poisson process equal to number of samples, the number of nuclei set at three times the sample size, and uncertainty of coordinate allowed to vary for 4 km and 8 km based on their movements rates described in the literature (Hegrenes 1992; Guillot et al. 2005; Donabauer 2009). After the most likely k was determined, an additional run with that particular k was run to determine membership probabilities with the same parameters. A PCA was performed with the package adegenet (Jombart 2008; Jombart and Ahmed 2011) in R (R Core Team 2016) to summarize genetic diversity among our samples.

RESULTS

Null alleles were identified at all 10 loci and all seven sites for Channel Catfish, which were corrected for prior to analysis. However, for every case the total number of false homozygotes observed was low. After correcting for null alleles, only three tests of 70 possible tests (10 loci with seven sites) deviated from HWE after nonsequential Bonferroni correction. No deviations from linkage equilibrium were detected. Allelic richness (A_R) ranged from 5.42 to 6.51, H_o ranged from 0.65 to 0.71, and H_e ranged from 0.63 to 0.75 (Table 1). Both H_o and H_e values were higher for the four sites on the Wabash River than for the Ohio River sites, although they were not significant (U-test: H_o, $P = 0.057$; H_e, $P = 0.114$); and the four sites on the Wabash River also had three of the highest A_R values. Overall average relatedness (r) among all individuals (mean ± SD = 0.057 ± 0.099) and all sites was low (Table 1).

Significant genetic differentiation was detected over all sites and loci ($\Phi_{ST} = 0.087$, $P < 0.001$), and a majority (18 of 21, 85.7%) of pairwise tests showed significant genetic differentiation (Φ_{ST} ranged from 0.009 to 0.184; Table 2). An isolation-by-distance pattern was detected with a significant Mantel's test between genetic [$\Phi_{ST}/(1 - \Phi_{ST})$] and geographic (rkm) distance (correlation coefficient $r = 0.20$, $P = 0.0291$; Figure 2).

Three groups were identified among all individuals using PCA, STRUCTURE, and GENELAND (Figures 3, 4). For STRUCTURE, the ln $\Pr(x|k)$ was maximized at $k = 3$, −4,042.70; the delta k was maximized at $k = 3$, 812.13. All fish in the analysis had q-values that were greater than 0.8 for the cluster to which they were assigned (Figure 3). GENELAND $k = 3$ had the highest average log posterior probability for each independent run; thus, the individual run at $k = 3$ was used to determine membership probabilities (Figure 3). The observed groups in the PCA, STRUCTURE, and GENELAND followed a geographic pattern; one group predominantly occurred in the northern extreme, two groups occurred in the southern extreme, and the central sites consisted of mixtures.

The assignment of individuals to groups produced a pattern consistent with adult Channel Catfish comprising mixtures. Both PCA and STRUCTURE were consistent (i.e., the same individuals clustered into the same group). Group 1 consisted of individuals from the three upstream Wabash River sites (WA_268, WA_189, and WA_136), group 2 consisted of individuals from three Wabash River sites and two Ohio River sites (WA_189, WA_136, WA_19, OH_1511, and OH_1548), and group 3 consisted of individuals from two Wabash River sites and two Ohio River sites (WA_136, WA_19, OH_1478, and OH_1511; Figures 3, 4). Three sites had 100% of the individuals captured at that site assigned to a single group (WA_268

TABLE 2. Summary of pairwise tests of genetic differentiation among adult Channel Catfish from sites sampled on the Wabash (WA) and Ohio (OH) rivers; Φ_{ST} values are presented above the diagonal, and geographic distance (rkm) are presented below the diagonal. Significant pairwise differentiations (after Bonferroni correction) are denoted by an asterisk (*).

Site	WA_268	WA_189	WA_136	WA_19	OH_1478	OH_1511	OH_1548
WA_268		0.009	0.072*	0.123*	0.173*	0.157*	0.184*
WA_189	79		0.029*	0.080*	0.142*	0.102*	0.123*
WA_136	132	53		0.018	0.069*	0.049*	0.074*
WA_19	249	170	117		0.066*	0.050*	0.087*
OH_1478	379	299	246	129		0.021	0.172*
OH_1511	412	332	279	162	33		0.113*
OH_1548	449	369	316	199	70	37	

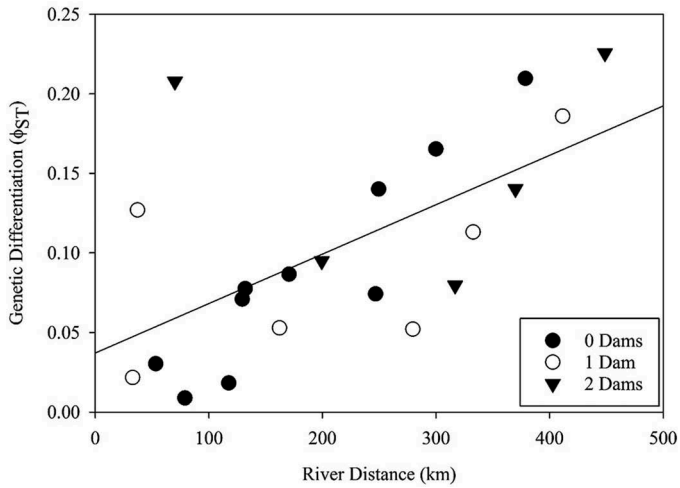

FIGURE 2. Scatter plot with of genetic differentiation [$\Phi_{ST}/(1 - \Phi_{ST})$] values among Channel Catfish versus river distance (rkm) among sites on the Wabash and Ohio rivers; black dots show pairwise comparisons with zero dams between sites, white dots show pairwise comparisons with one dam between sites, and black triangles show pairwise comparisons with two dams between sites. Mantel's test was significant showing a positive linear relationship (correlation coefficient $r = 0.20$, $P = 0.0291$).

to group 1, OH_1478 to group 2, and OH_1548 to group 3), whereas all other sites had individuals assigned to multiple groups (Figure 3). Lastly, all groupings showed high (relative

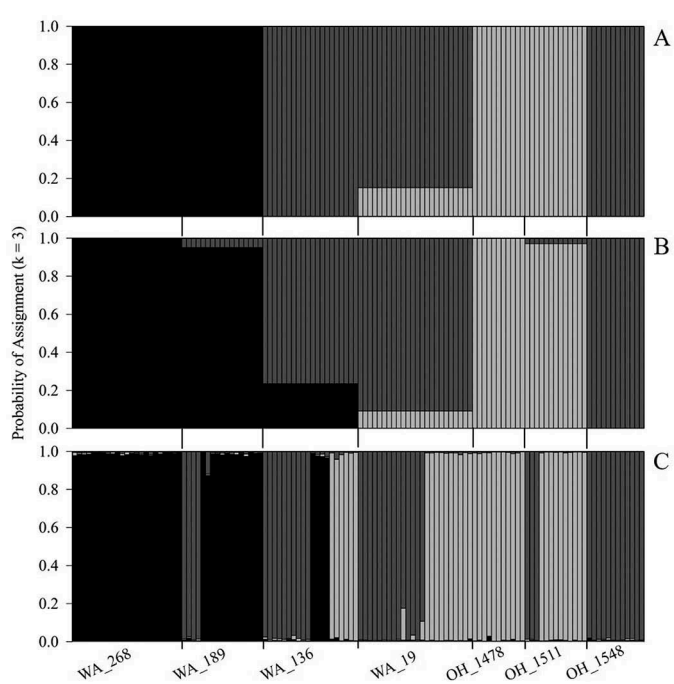

FIGURE 3. GENELAND and STRUCTURE plots for $k = 3$. (**A**) GENELAND plot with the geographic uncertainty set to 4 km, (**B**) GENELAND plot with the geographic uncertainty set to 8 km, (**C**) STRUCTURE plot. Each color corresponds to a different grouping: black is group 1, dark gray is group 2, and light gray is group 3.

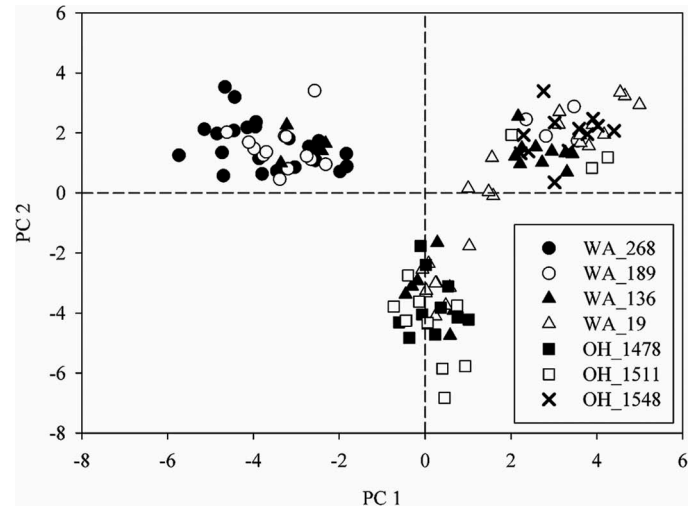

FIGURE 4. Principal components analysis of adult Channel Catfish from the Wabash and Ohio rivers; each point represents one fish.

to pairwise site comparisons) and significant genetic differentiation values (Φ_{ST}: group 1 − group 2: 0.156, group 1 − group 3: 0.128, group 2 − group 3: 0.108; $P < 0.001$).

For the GENELAND analysis with a geographic uncertainty value of 4 km, all individuals from WA_268 and WA_189 were assigned to group 1, all individuals from WA_136 and OH_1548 were assigned to group 2, and all individuals from OH_1478 and OH_1511 were assigned to group 3. Individuals from WA_19 had mixed ancestry according to this model, sharing ancestry with both groups 2 and 3, although all individuals had a majority of their ancestry assigned to group 2 (Figure 3). Additionally, there is weak evidence of isolation by distance between WA_19 and OH_1478. Similar patterns were found when GENELAND was run with geographic uncertainty set at 8 km with regard to groupings, yet there is a clearer pattern of isolation by distance due to some changes in probabilities of assignment (Figure 3). This can be seen with the addition of some ancestry from group 1 into WA_136, group 2 into WA_189, and group 2 into OH_1511.

From the Ohio River, three pairwise comparisons had at least one artificial impoundment between sites (OH_1478 versus OH_1511, OH_1511 versus OH_1548 with one dam, OH_1478 versus OH_1548 with two impoundments). The comparisons with two impoundments between sites had the second highest observed Φ_{ST} with the fourth smallest distance between sites (Table 2, Figure 1). Results were inconsistent for the sites with one dam in between them. Site OH_1511 was differentiated from OH_1548; these sites were the second closest together but had the seventh highest Φ_{ST} (Table 2, Figure 1). However, OH_1478 was not differentiated from OH_1511 (Table 2). Further, the three sites in proximity of an artificial impoundment had the lowest H_o and three of the four lowest H_e values.

DISCUSSION

Genetic Structuring

Channel Catfish from the Wabash and Ohio rivers showed genetic structuring that followed an isolation-by-distance pattern, which is common in riverine fishes (Hansen and Mensberg 1998; Pogson et al. 2001; Maes and Volckaert 2002; Primmer et al. 2006; Griffiths et al. 2009). Additionally, estimates of Φ_{ST} clearly differentiated Channel Catfish from the various collection locales, which could be explained by geographic distance. This pattern of isolation by distance is consistent with previous work done on Channel Catfish migration, which detected occasional long-distance movements with short-distance movements more frequent. This result of isolation by distance is particularly interesting given the presence of population sizes large enough to support commercial harvest.

Previous studies found that Channel Catfish will migrate, sometimes over 100 km, into smaller rivers and tributaries to find suitable spawning habitat in the spring but will return to the main stem in the fall (Dames et al. 1989; Newcomb 1989; Pellett et al. 1998; Fago 1999; Wendel and Kelsch 1999). However, in past studies a majority of Channel Catfish, including one on the Wabash River, were found to move fewer than 16 km and had small summer home ranges (approximately 5.7 km: Hegrenes 1992; Wendel and Kelsch 1999; Donabauer 2009). We took this into consideration when analyzing their genetic structure with GENELAND. We ran the analysis twice, once with the geographic uncertainty set to 4 km and again with 8 km. The resulting plots are similar, although an isolation-by-distance pattern does start to appear in the 8-km plot in mid-river Wabash sites. Thus, due to the nature of their movements and based on the observed isolation-by-distance pattern, gene flow is more likely to occur between sites that are closer together, yet some longer distance gene flow is probable.

The presence of large-scale movements is also consistent with the migratory nature of Channel Catfish in other systems (Pellett et al. 1998; Wendel and Kelsch 1999). Both PCA and STRUCTURE analysis revealed several individuals that were captured at one site, yet grouped with individuals captured from other sites. This indicates that the genetic characteristics of Channel Catfish in the Wabash and Ohio rivers were consistent with adults moving long distances from the reproductive sites, which has been demonstrated in other migratory fishes (Shaklee et al. 1999; Dammannagoda et al. 2011). In Channel Catfish in the Wabash and Ohio rivers, isolation by distance is consistent with long-distance movements because larger adult fish exhibit homing tendencies (Pellett et al. 1998). Thus, the genetic data indicate that Channel Catfish may exhibit reproductive isolation and natal homing in the Wabash and Ohio rivers even though they are capable of undertaking long-distance movements. When comparing fish by their assigned groupings by both PCA and STRUCTURE, high differentiation values were found, indicating additional support for the mixed-population structure. These differentiation values were higher than half (~54%) of the pairwise site comparisons, indicating substantial differentiation relative to the locations where these fish were captured. The pairwise site comparisons that had higher differentiation values involved sites that were far apart (e.g., upstream Wabash and Ohio river sites) or between the Ohio River sites separated by L&D 52.

Impact of Dams

On the Ohio River, artificial impoundments affected genetic diversity of Channel Catfish relative to Channel Catfish from the Wabash River. Channel Catfish from the Ohio River sites had lower heterozygosity levels than those from the Wabash River, which may be the result of the impoundments on the Ohio River. Impoundments are known to cause a loss of genetic diversity in anadromous steelhead *Oncorhynchus mykiss* that were isolated above impoundments in California (Nielsen et al. 1997). Additionally, in a more general sense, impoundments blocking migratory routes, and the conversion of free-flowing rivers to reservoirs have led to the decline of numerous fish species in Illinois (Santucci et al. 2005). Thus, fragmentation of populations can lead to a decrease in gene flow and an increase in genetic drift and inbreeding, which can eventually lead to a loss of rare alleles, reducing total genetic variability (Hedrick and Miller 1992; Hutchison and Templeton 1999).

The impoundments also had an impact with regard to the genetic structure of the Channel Catfish population in the Ohio River. Three pairwise site comparisons (OH_1478 and OH_1548, OH_1511 and OH_1548, and OH_1478 and OH_1511) had barriers between them in the form of locks and dams and were also the three of the four geographically closest, all within 70 rkm (average distance between sites, 215.10 rkm). Interestingly, two of these comparisons (OH_1478 and OH_1548, OH_1511 and OH_1548) had high genetic differentiation, yet the comparison between sites OH_1478 and OH_1511, which is the geographically closest comparison (33 rkm), lacked significant genetic differentiation. This could be due to the nature and ages of the dams; L&D 52 has been in place ~50 years longer than SM L&D; thus, it has had longer to affect the genetic structuring of Channel Catfish in the Ohio River.

Several studies have found high levels of differentiation in impounded riverine systems (Yamamoto et al. 2004; Hanfling and Weetman 2006), while some show low levels of differentiation (Neraas and Spruell 2001; Denier et al. 2007; Reid et al. 2008). The two comparisons not conforming to the observed isolation-by-distance pattern were separated by L&D 52. This likely means that either L&D 52 is a stronger barrier to movements than SM L&D, thus driving up differentiation and indicating there may be gene flow

(movements) over SM L&D between fish immediately surrounding the dams, or SM L&D has not yet been in place long enough to alter their genetic differentiation. The resulting amount of differentiation between separated populations depends upon the reduction of gene flow, effective population size, and an increase in genetic drift in isolated populations (Jager et al. 2001; Meldgaard et al. 2003; Yamamoto et al. 2004; Hanfling and Weetman 2006; Reid et al. 2008). Overall though, this indicates that impoundments may have complex effects on the genetic structure of Channel Catfish and fishes in general.

Past studies on Channel Catfish movement in systems with artificial impoundments have generated mixed results; some found Channel Catfish are able to pass over low-head dams and through locks and dams (Muncy 1958; Wendel and Kelsch 1999; Wilcox et al. 2004), yet others found low-head dams were strong barriers to movement (Welker 1967; Gerhardt and Hubert 1990; Butler and Wahl 2011). Therefore, the effect of impoundments on Channel Catfish genetic structure and diversity likely depends on the system and the types of impoundments (e.g., low-head, lock and dam) in that particular system. Therefore, determining the effects of impoundments on Channel Catfish will require additional data and more intensive sampling on Channel Catfish in highly impounded systems. Using more loci and more individuals in future studies could reveal fine-scale differences among locations and populations that were not detected by the current study. These data could provide additional pertinent information for the management of Channel Catfish in Midwestern rivers.

Management Implications

Since Channel Catfish are one of the most sought-after fishes in large Midwestern rivers, their management is of utmost importance. We found lower genetic diversity indices and differentiation values that did not conform to the pattern of isolation by distance found in all other pairwise comparisons in fishes immediately upstream or downstream from dams. Therefore, it is likely that dams are affecting these fish relative to the fish in the free-flowing Wabash River. If dams restrict Channel Catfish movement, which appears likely, this could lead to a long-term loss of genetic diversity due to an increase in the impacts of genetic drift. Thus, careful consideration should be taken for the long-term health of these populations to ensure they remain genetically viable for sustainable harvest.

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